



Quantum Column Generation for Graph Coloring

An end-to-end tutorial: from a 5G frequency-assignment problem to Dirac-3 hardware

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Outline

- 1 The problem
- 2 Column generation
- 3 Dirac-3 and Motzkin–Straus
- 4 Antenna demo
- 5 Scaling and outlook

Frequency assignment is graph coloring

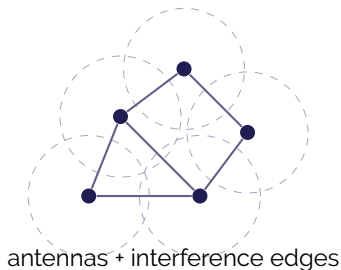
A wireless network has n antennas. Two antennas whose coverage footprints overlap **interfere** when they share a frequency.

Problem: assign frequencies so that no two interfering antennas share one, using as few frequencies as possible.

Build the **interference graph**:

- one vertex per antenna,
- an edge between any two antennas closer than the interference radius.

Minimum frequencies = $\chi(G)$, the chromatic number.



Why this is hard

Vertex coloring is NP-hard

No polynomial-time exact algorithm is known. Direct MILP works on graphs of ~ 50 vertices; beyond that, the formulation explodes.

Practical approaches:

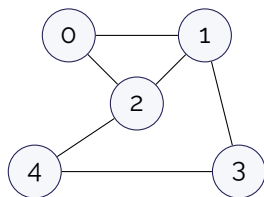
- **Greedy** (DSATUR, LF): fast, no quality guarantee, often off by 30–50% on dense graphs.
- **Direct MILP**: optimal up to ~ 50 nodes, then intractable.
- **Column generation**: scales to thousands of nodes by generating only the columns we need.

Column generation in one picture



- Start with trivial columns (one IS per vertex).
- Repeat: solve RMP, get duals, solve PSP for a profitable column, add it.
- Stop when no column has positive reduced cost. Solve the final ILP.

A worked example: the 5-vertex paper graph



$$n = 5, m = 6, \chi = 3$$

Classical CG run (from notebook):

- Iter 1: RMP obj = 5 PSP finds $\{0, 3\}$
- Iter 2: RMP obj = 3.5 PSP finds $\{1, 4\}$
- Iter 3: RMP obj = 3 PSP finds no new column

Final ILP picks three ISs: $\{0, 3\}, \{1, 4\}, \{2\}$

$$\Rightarrow \chi = 3.$$

Three CG iterations close an exponential-column LP.

Where classical CG slows down

- Each CG iteration solves a **max-weight independent set** (NP-hard).
- Classical MWIS oracles (MILP) return **one** IS per call.
- Many iterations $\Rightarrow$ many sequential PSP solves.

Two levers for speed

- 1 Solve each PSP faster.
- 2 Return multiple ISs per PSP solve (richer column pool, fewer iterations).

Dirac-3 attacks both at once.

Motzkin–Straus: clique-finding as a continuous QP

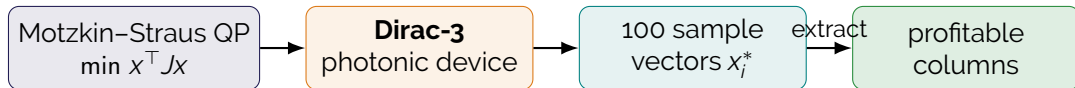
For an undirected graph G with adjacency A and stability number $\alpha(G)$:

$$\frac{1}{2} \left(1 - \frac{1}{\alpha(G)} \right) = \max_{\substack{\mathbf{1}^\top \mathbf{x} = 1 \\ \mathbf{x} \geq 0}} \frac{1}{2} \mathbf{x}^\top A \mathbf{x}.$$

Maximisers are supported on **maximum cliques**.

- Run Motzkin–Straus on the *complement* $\bar{G} \implies$ recover MIS in G .
- For weighted MIS: use Gibbons' weighted variant with dual matrix B .

Dirac-3 entropy computer: native QP sampler



- **Single device call:** 100 samples, each near a different local optimum on the simplex.
- Each sample is post-processed (support-threshold + pruning + local search) into an independent set.
- One device call → typically **5–30 distinct ISs**.

Quantum CG: one algorithm box

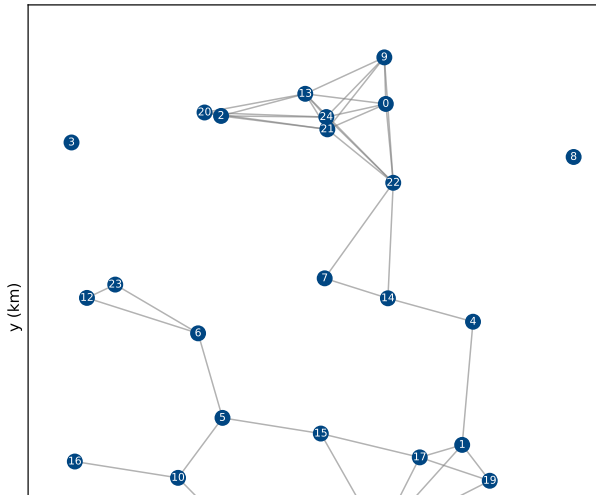
Algorithm: quantum column generation

- 1 Initialise column pool with singleton ISs.
- 2 **Repeat:**
 - Solve RMP $\rightarrow$ get dual prices y_v .
 - Submit *weighted* Motzkin–Straus QP to Dirac-3.
 - Receive 100 samples; extract all profitable ISs.
 - Add unique new ISs to the column pool.
 - Stop if no new profitable column was found.
- 3 Solve final integer master $\rightarrow$ optimal coloring on the pool.

Same algorithm as classical CG — only the PSP oracle is swapped.

Synthetic antenna network (25 antennas in 10 km box)

Synthetic antenna network — 25 antennas, interference radius = 2.5 km

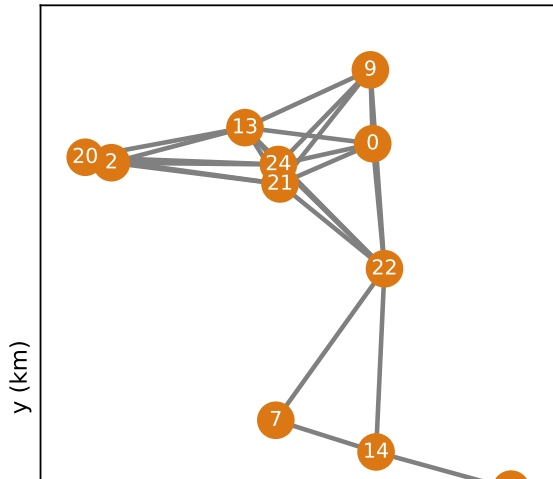


- 25 antennas, uniform in 10 km box (rng seed 7).
- Interference radius 2.5 km $\Rightarrow$ 44 edges.
- BFS from the highest-degree antenna picks a connected 12-vertex subgraph that we use as the head-to-head test case on the next slide.

All three solvers run on the same induced subgraph; the visualisation below shows their colorings side by side.

The 12-antenna subgraph

12-antenna BFS subgraph (12 nodes, 27 edges)



- 12 antennas, 27 interference edges.
- Dense enough that greedy heuristics over-allocate.
- Small enough that direct MILP closes optimally in tens of milliseconds.
- Big enough that the column-generation pricing subproblem is non-trivial: 25–30 maximal ISs.

Three solvers, one subgraph

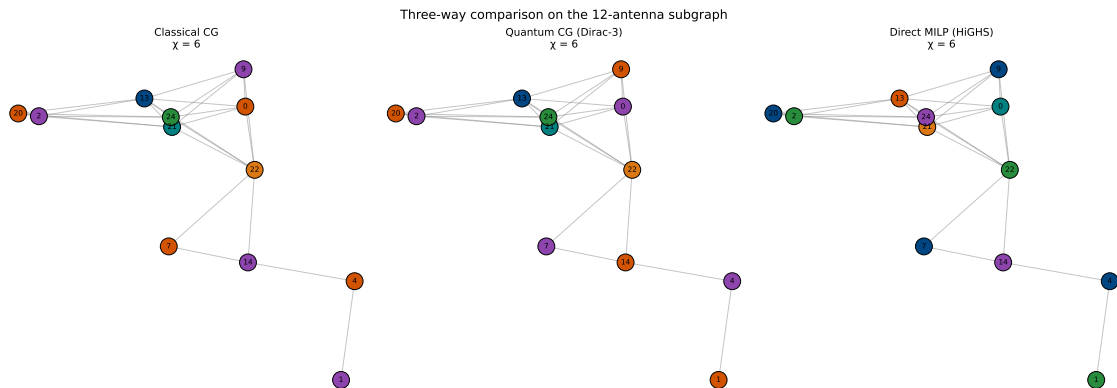
Classical CG
MILP MWIS oracle
one IS / call
fast for small n

Quantum CG
Dirac-3 MWIS oracle
many ISs / call
hardware bound

Direct MILP
exact, optimal
scipy / Hexaly
cliff above ~ 50 nodes

All three run on the same 12-vertex subgraph. The next slide overlays each method's coloring on the antenna layout.

Side-by-side coloring



All three methods find the proven optimum $\chi = 6$. They assign different vertices to different colors (the labelling is arbitrary) but every coloring is valid and uses 6 frequencies.

Head-to-head on the 12-antenna subgraph

Method	χ	runtime (s)	cols / call	notes
Classical CG	6	0.02	9 – 12	heuristic
Quantum CG (Dirac-3)	6	50.8	15 – 25	heuristic, live device
Direct MILP	6	0.04	—	✓ optimal

All three find the optimal $\chi = 6$ on this instance. Quantum CG uses *fewer iterations* (richer column pool per call) at the cost of higher per-call latency. Direct MILP wins on wall-time *at this size*; CG wins at scale.

Where does Dirac-3 win?

Sweet spot: dense graphs, $n \in [50, 300]$

- Direct MILP runs out of memory / time.
- Classical CG works but needs many sequential PSP solves.
- Quantum CG returns 2–4× more columns per call; converges in **half the iterations** of classical CG on $n = 100$ benchmark instances.

Limits

- Per-call latency on Dirac-3 cloud is ~30–90 s. Wall-clock advantage requires problems where each iteration is itself slow (large MWIS subproblems).
- For tiny graphs ($n < 20$), direct MILP wins outright.

Key takeaways

- Quantum column generation **drops into the same algorithm** as classical CG — swap one subroutine.
- **Dirac-3 is a clique sampler**: one device call yields many independent sets, which is exactly what CG needs.
- On the synthetic 5G antenna instance all three approaches found the optimal coloring; the quantum path wins on *columns per call* and *iterations to converge*.
- Same algorithm scales naturally: notebooks 02b / 03b run live on user graphs; replayed bundled data lets colleagues reproduce our results without device access.

Notebook: `notebooks/04_tutorial_end_to_end.ipynb`

Deep dive: `notebooks/slides/deep_dive/deep_dive.pdf`

References: arXiv:2301.02637 (Coelho et al.)